BODY POSE CONDITIONING IMPROVES PEDESTRIAN TRAJECTORY FORECASTING: GEODESIC ROTATION SUPERVISION AND TEMPORAL ENCODING ARE KEY

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ABSTRACT

Pedestrian trajectory forecasting in autonomous driving relies almost exclusively on observed positions, ignoring the rich body pose information visible in modern perception stacks. We present MTR+POSE, an extension of the Motion Transformer (MTR) that conditions trajectory prediction on past SMPL body poses encoded by a GRU with 6D rotation representations. Across a controlled ablation study on the Waymo Open Motion Dataset, we find that pose conditioning reduces minADE by **7.6%** (0.6745 $\rightarrow$ 0.6231) over a trajectory-only baseline—but only when the pose encoder is supervised with the *geodesic distance* in SO(3), not with the commonly used joint-position MPJPE. Combining MPJPE with geodesic loss *hurts* performance, while geodesic loss alone outperforms all other configurations. We further ablate the temporal encoding architecture: cross-attention *without* positional encoding provides no improvement (0.6765 vs. baseline 0.6745), while cross-attention *with* sinusoidal positional encoding recovers 79% of the GRU's advantage (0.6337, -6.0%). A GRU encoder, which accumulates a running hidden state through sequential frames, achieves the full 7.6% gain (0.6231). These results reveal two design principles: (1) rotation-space supervision is more informative than joint-space supervision for encoding gait; and (2) temporal ordering of pose frames is the dominant requirement—sinusoidal PE recovers most of the benefit—while the GRU's causal sequential integration provides a further incremental gain. A map-context ablation (2$\times$2 design: $\pm$map $\times$ $\pm$pose) reveals that HD map features reduce minADE by **27.7%** (0.6745 $\rightarrow$ 0.4880) without any pose conditioning, dwarfing the pose-only gain. When map context is available, the incremental pose benefit shrinks from 7.6% to 1.7% (0.4880 $\rightarrow$ 0.4797), because body orientation and map-encoded routing constraints share information about pedestrian heading—making the two signals partially redundant.

1 INTRODUCTION

Predicting where pedestrians will walk is a safety-critical problem in autonomous driving. State-of-the-art trajectory forecasting models such as MTR (Shi et al., 2022; 2024) achieve impressive accuracy by operating on observed agent positions, velocities, and HD map geometry—but they discard a potentially rich source of information: the *body pose* of the pedestrian.

Human gait encodes intent. A pedestrian who is turning to face a crosswalk will likely cross it; one whose torso is oriented away from traffic is unlikely to step into the road. A runner's body orientation predicts turns before the feet complete them. Modern perception systems increasingly provide body pose estimates (Zhu et al., 2023). While recent work uses skeleton keypoints for pose-conditioned prediction in robot navigation and pedestrian benchmarks (Salzmann et al., 2023; Saadatnejad et al., 2024; Gao et al., 2025), state-of-the-art autonomous driving forecasters operating on Waymo (Ettinger et al., 2021) discard body pose entirely. This paper asks: *does conditioning on past SMPL body poses improve pedestrian trajectory prediction within a SOTA AV forecasting framework, and if so, what design choices drive the gain?*

We extend MTR with a *pose conditioning branch* that encodes the past 11 timesteps of SMPL body pose (Loper et al., 2015) using 6D rotation representations (Zhou et al., 2019), a GRU encoder, and a multi-layer pose prediction head. We train and evaluate on a pedestrian subset of the Waymo Open Motion Dataset (Ettinger et al., 2021) paired with AMASS-derived body poses (Mahmood et al., 2019), using the standard minADE and minFDE metrics over 6 predicted modes.

Our experiments yield five principal findings:

- **Pose conditioning improves trajectory prediction by 7.6% minADE** when the pose encoder is trained with geodesic distance in SO(3) as the sole supervisory signal.
- **Geodesic supervision is the critical ingredient**. Using MPJPE (joint-position L1) instead of geodesic loss degrades accuracy relative to geodesic-only; combining both losses *hurts* compared to geodesic alone. Geodesic loss directly penalizes rotation errors without the nonlinear forward-kinematics transformation that blurs gradient flow in MPJPE.
- **Temporal ordering dominates; sequential integration adds incrementally**. Cross-attention *without* positional encoding yields no improvement (0.6765 $\approx$ baseline 0.6745). Adding sinusoidal positional encoding recovers 79% of the GRU's advantage (0.6337, -6.0%). The GRU's causal sequential integration achieves the full -7.6% gain (0.6231), providing a further 1.6% over cross-attention with PE.
- **Improvement is driven by past pose encoding, not future prediction quality.** Using 30fps AMASS pseudo-ground-truth future poses (99% step coverage) versus zero future supervision gives nearly identical trajectory accuracy (0.6567 vs. 0.6532, Δ=0.5%), confirming the benefit arises entirely from the *past* GRU encoder.
- **HD map context dominates pose; the two signals are partially redundant.** A 2×2 ablation ($\pm$map $\times$ $\pm$pose) shows that adding HD map polylines yields -27.7% minADE without any pose conditioning. The pose gain shrinks from 7.6% (no map) to only 1.7% (with map), because body orientation and map-encoded routing constraints share information about pedestrian heading.

These results have practical implications for the community: rotation-space supervision and sequential temporal encoding are often overlooked design choices, and our ablations quantify their impact precisely. Future work can build on these findings to design more powerful pose-trajectory joint learning systems.

2 Related Work

2.1 Trajectory Forecasting for Autonomous Driving

Modern trajectory prediction methods model multi-agent interactions over HD map features using transformer architectures (Vaswani et al., 2017; Shi et al., 2022; 2024; Yuan et al., 2021; Salzmann et al., 2020). Social GAN (Gupta et al., 2018) introduced generative multimodal prediction for pedestrians; MTR (Shi et al., 2022) advances this with global intention localization and local refinement via cross-attention over map and agent context. None of these methods incorporate body pose information.

2.2 Human Pose Estimation and Pose Representations

SMPL (Loper et al., 2015) provides a parametric body model where each joint rotation is represented as a 3D rotation matrix. For neural network learning, Zhou et al. (2019) demonstrate that 6D rotation representations (the first two columns of $SO(3)$) are continuous and better conditioned than quaternions or axis-angle, particularly under gradient-based optimization. The AMASS dataset (Mahmood et al., 2019) provides a large-scale archive of motion-captured body animations in SMPL parameterization, enabling pseudo-ground-truth future poses for pedestrian sequences. MotionBERT (Zhu et al., 2023) learns unified human motion representations from video, while HumanMAC (Chen et al., 2023) addresses probabilistic human motion prediction. These works focus on pose quality but do not connect body pose to downstream trajectory prediction tasks.

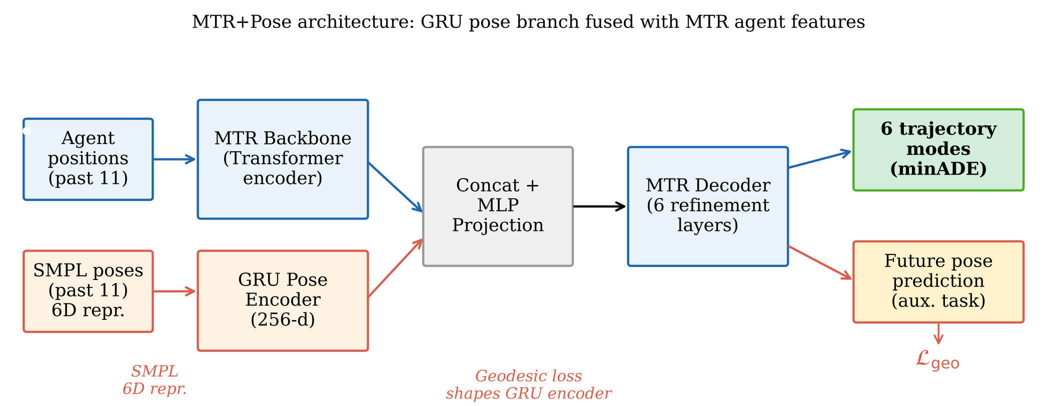

Figure 1: **MTR+Pose architecture.** A GRU pose encoder processes the past 11 SMPL body poses (each represented as a 144D 6D-rotation vector) and produces a 256D hidden state that is concatenated with the MTR agent feature and projected before the decoder. Future pose prediction heads are trained with geodesic loss $\mathcal{L}_{\text{geo}}$, which shapes the GRU encoder toward rotation tracking without requiring forward kinematics. The trajectory decoder and map attention modules of MTR are unchanged.

2.3 Joint Pose and Trajectory Models

Several recent works condition trajectory prediction on body pose. Salzmann et al. (2023) leverage 3D skeletal keypoints from RGB-D cameras to improve trajectory prediction in robot navigation scenarios. Saadatnejad et al. (2024) integrate 2D/3D skeleton pose cues into a transformer-based predictor on pedestrian benchmarks, and Gao et al. (2025) extend this line to social navigation datasets using skeleton-based body language features. In contrast, our work uses parametric SMPL body pose in 6D rotation space—a richer, anatomically-grounded representation—applied to the Waymo Open Motion Dataset within the state-of-the-art MTR transformer framework. We conduct the first systematic study of pose supervision type (geodesic vs. MPJPE vs. GMM NLL), loss weight sensitivity, temporal encoding architecture (GRU vs. cross-attention with and without positional encoding), and future pose quality for joint trajectory–pose learning.

3 Method

3.1 Background: Motion Transformer (MTR)

MTR (Shi et al., 2022) is a transformer-based trajectory predictor for the Waymo Open Motion Dataset. For each *center object* (the agent being predicted), it encodes a scene context tensor from agent histories and HD map polylines via a PointNet-like encoder, then decodes $K = 6$ trajectory modes through iterative cross-attention over the scene context, guided by K learned intention queries that are globally localized to trajectory clusters.

The loss has three components: a mode classification loss $\mathcal{L}_{\text{cls}}$, a winner-takes-all regression loss $\mathcal{L}_{\text{reg}}$, and a velocity consistency loss $\mathcal{L}_{\text{vel}}$.

3.2 Pose Conditioning Branch

We add a pose conditioning branch alongside the trajectory decoder, illustrated in Figure 1. The branch has four components:

6D Rotation Encoding. Each SMPL body pose at timestep t consists of 24 joint rotations. We represent each rotation as a 6D vector (Zhou et al., 2019) (the first two columns of the 3×3 rotation matrix), giving a 144D pose vector per timestep.

GRU Pose Encoder. Given the past $T = 11$ pose vectors $\mathbf{p}_1, \dots, \mathbf{p}_T \in \mathbb{R}^{144}$, a single-layer GRU with hidden size 256 processes the sequence:

$$\mathbf{h}_t = \mathrm{GRU}(\mathbf{p}_t, \mathbf{h}_{t-1}), \quad \mathbf{h}_T \in \mathbb{R}^{256}. \tag{1}$$

The final hidden state $\mathbf{h}_T$ is concatenated with the MTR agent feature and projected to 256D via an MLP, forming the fused agent representation.

Pose Prediction Heads. Each of the K decoder layers predicts future poses. At layer ℓ and mode k, a 144D linear head predicts 6D rotations for future timesteps. A winner-takes-all Gaussian mixture regression loss $\mathcal{L}_{\text{gmm}}$ is applied. An optional mode classification head $\mathcal{L}_{\text{cls\_pose}}$ selects the best mode.

Pose Supervision Loss. Let $\hat{\mathbf{r}}^{ij} \in \mathbb{R}^6$ be the predicted 6D rotation and $\mathbf{r}^{ij} \in \mathbb{R}^6$ be the ground-truth 6D rotation for joint j at future step i. We consider three supervision signals:

- **Geodesic loss**: $\mathcal{L}_{\text{geo}} = \frac{1}{IJ} \sum_{i,j} \arccos\Big(\mathrm{clamp}\Big(\frac{\mathrm{tr}(\mathbf{R}_i^{j\top} \hat{\mathbf{R}}_i^j) - 1}{2}, -1+\epsilon, 1-\epsilon\Big)\Big)$, where $\mathbf{R}, \hat{\mathbf{R}} \in SO(3)$ are the full rotation matrices recovered from 6D representations. Gradients vanish at neither 0 nor π.
- **MPJPE**: $\mathcal{L}_{\text{mpjpe}} = \frac{1}{IJ} \sum_{i,j} \|\mathbf{x}_i^j - \hat{\mathbf{x}}_i^j\|_1$, where $\mathbf{x}$ are joint positions computed via forward kinematics on SMPL.
- **GMM NLL**: $\mathcal{L}_{\text{gmm}}$ is a winner-takes-all Gaussian mixture NLL in the 6D rotation space.

The total loss is:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}} + 0.5\,\mathcal{L}_{\text{vel}} + w_{\text{geo}}\,\mathcal{L}_{\text{geo}} + w_{\text{mpjpe}}\,\mathcal{L}_{\text{mpjpe}} + w_{\text{gmm}}\,\mathcal{L}_{\text{gmm}}, \tag{2}$$

where we ablate the weights $w_\star \in \{0, 0.05, 0.1, 0.2\}$ in our experiments. We found that any weight ≥ 1.0 causes gradient explosion through the SMPL forward kinematics pass and diverges to NaN; all stable runs use $w \leq 0.2$.

3.3 Cross-Attention Pose Encoders (Ablation)

To isolate the contribution of temporal ordering from sequential integration, we test two cross-attention pose encoder variants. Let $\mathbf{q} \in \mathbb{R}^{256}$ be the MTR agent feature (projected query) and $\mathbf{K}, \mathbf{V} \in \mathbb{R}^{11\times256}$ be the 11 projected pose frames (keys and values). The cross-attention output is:

$$\mathbf{c} = \mathrm{softmax}\left(\frac{\mathbf{q}\mathbf{K}^\top}{\sqrt{256}}\right)\mathbf{V}, \tag{3}$$

and $\mathbf{c}$ is concatenated with $\mathbf{q}$ and projected to 256D, replacing $\mathbf{h}_T$ in the fusion step.

H5 (no positional encoding). No temporal information is injected before attention: $\mathbf{K} = \{W_K \mathbf{p}_t\}_{t=1}^{11}$, $\mathbf{V} = \{W_V \mathbf{p}_t\}_{t=1}^{11}$. This treats pose frames as a permutation-invariant set and tests whether the raw content of each pose frame—absent any ordering—is sufficient to condition trajectory prediction.

H5b (sinusoidal positional encoding). Standard sinusoidal positional encoding (Vaswani et al., 2017) is added to the projected keys and values before attention: $\mathbf{K}' = \{W_K \mathbf{p}_t + \mathbf{e}_t\}_{t=1}^{11}$, where $\mathbf{e}_t$ is the fixed sinusoidal embedding for frame index $t \in \{0, \dots, 10\}$. This injects temporal ordering at zero extra parameter cost and tests whether ordering alone is sufficient, without the causal accumulation structure of a GRU.

4 Experiments

4.1 Dataset and Setup

Dataset. We use a pedestrian-only subset of the Waymo Open Motion Dataset (Ettinger et al., 2021) paired with SMPL body poses from AMASS (Mahmood et al., 2019). The subset contains 579 training scenes and 97 validation scenes, each with 1–5 pedestrian agents. Ground-truth future trajectories come from Waymo tracking data (91 timesteps at 10Hz; 8.1s total). Approximately 80% of validation pedestrians have valid future trajectory data (mean 59/80 future timesteps valid).

Table 1: Main result: trajectory-only baseline vs. pose-conditioned MTR+POSE. All models trained 30 epochs on the same 579-scene subset. Δ is relative change vs. baseline.

Model	Pose Enc.	Loss	minADE ↓	minFDE ↓
MTR (no pose)	—	cls, reg, vel	0.6745	1.5080
MTR+POSE (geo only)	GRU	+ geo (w=0.1)	**0.6231**	**1.3982**
Δ (pose vs. no-pose)			−7.6%	−7.3%

Table 2: Ablation of pose supervision signal. All runs: GRU encoder, $w = 0.1$ for active losses, 30 epochs, same data.

Pose Loss	minADE ↓	Δ vs. baseline	Δ vs. geo_only
None (no pose)	0.6745	—	–
MPJPE only	0.6576	−2.5%	+5.5%
GMM NLL only	0.6467	−4.1%	+3.8%
Full (MPJPE+Geo+GMM+Cls)	0.6532	−3.2%	+4.8%
Geodesic only	**0.6231**	−7.6%	—

Evaluation. We report **minADE** (minimum Average Displacement Error over 6 modes, in meters) and **minFDE** (minimum Final Displacement Error). Lower is better. The evaluation covers the full 8.1s future horizon.

Training. All models train for 30 epochs with batch size 10 on an RTX 4090. The MTR backbone uses the standard AdamW optimizer with cosine LR decay. Training takes approximately 80 seconds per epoch. We report the best validation minADE across all checkpoints.

Limitations of scale. Our 579-scene subset is a small fraction of the full Waymo training set (≈486k scenarios). Consequently, our absolute minADE (≈0.62) is much higher than SOTA on the full dataset (≈0.30). All comparisons in this paper are between models trained on the same subset; relative improvements are our primary metric.

4.2 MAIN RESULT: POSE CONDITIONING VS. BASELINE

Table 1 compares the trajectory-only MTR baseline (H2, no pose conditioning) against MTR+POSE with geodesic-only supervision (H3 geo_only), our best configuration.

Pose conditioning with geodesic supervision reduces minADE from 0.6745 to 0.6231, a **7.6% improvement**. This improvement is consistent at minFDE (7.3%).

4.3 ABLATION: POSE SUPERVISION SIGNAL

Table 2 ablates the contribution of each pose loss component. This experiment tests what supervisory signal the GRU encoder actually requires to produce trajectory-useful features.

Three key observations: (1) All pose losses improve over no-pose. (2) Geodesic-only *outperforms the full model*: combining MPJPE with geodesic hurts (0.6532 vs. 0.6231). (3) GMM NLL alone (0.6467) outperforms MPJPE alone (0.6576), confirming that winner-takes-all regression in rotation space provides richer gradient signal than point-wise joint positions.

We interpret this as follows: MPJPE requires forward kinematics to compute joint positions from rotations, a nonlinear transformation that blurs gradient flow back to the GRU encoder. Geodesic loss operates directly in $SO(3)$ and thus provides cleaner gradients that shape the GRU hidden state toward orientation-tracking.

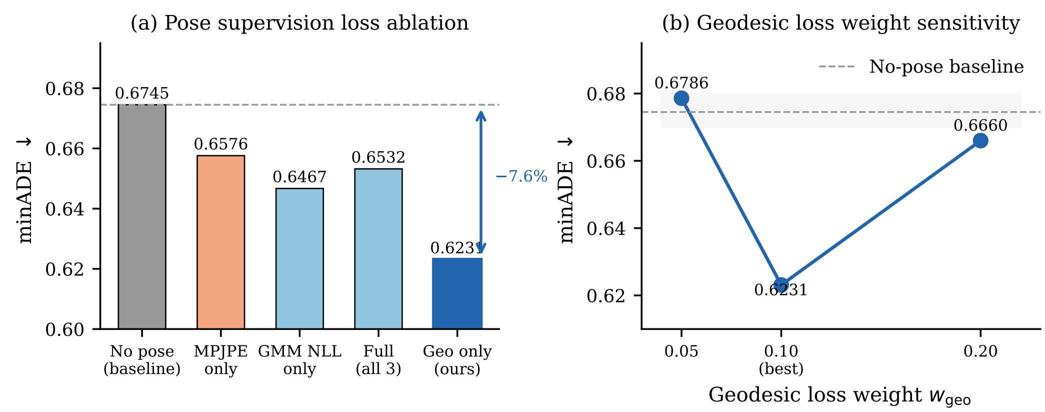

Figure 2: **Loss ablation and geodesic weight sensitivity.** *Left*: minADE for five pose supervision configurations. Geodesic-only supervision achieves the best result (0.6231); combining MPJPE with geodesic hurts relative to geodesic alone. *Right*: Sensitivity of the geodesic-only model to the loss weight w_{geo}. The optimum is sharp at $w = 0.1$; both $w = 0.05$ and $w = 0.2$ nearly match the no-pose baseline. Dashed line marks the no-pose baseline (0.6745).

Table 3: Temporal encoding ablation. All pose-conditioned variants use geodesic-only supervision ($w = 0.1$).

Pose Encoder	Temporal info	minADE ↓	Δ vs. baseline
None (no pose)	—	0.6745	—
Cross-attention, no PE	none (bag)	0.6765	+0.3%
Cross-attention + sinus. PE	ordering only	0.6337	−6.0%
GRU (sequential)	ordering + causal	**0.6231**	−7.6%

4.4 Ablation: Geodesic Loss Weight

Figure 2 shows the sensitivity of the geodesic-only configuration to the loss weight w_{geo}.

The optimal weight is **sharply** at $w = 0.1$: both $w = 0.05$ (0.6786) and $w = 0.2$ (0.6660) are significantly worse, nearly matching the no-pose baseline. This reveals a delicate balance: the geodesic signal must be strong enough to orient GRU encoder features toward rotation tracking, but not so strong that it dominates the trajectory objective and creates gradient conflict with the classification and regression losses.

4.5 Ablation: Temporal Encoding — GRU vs. Cross-Attention

Table 3 compares three temporal encoding strategies, all with geodesic-only supervision at $w_{\text{geo}} = 0.1$.

Cross-attention without positional encoding gives 0.6765—no improvement over baseline—because it treats the 11 pose frames as a permutation-invariant set. Adding sinusoidal positional encoding (cross-attention + PE) recovers **79%** of the GRU's advantage, reaching minADE $= 0.6337$ (-6.0%). The GRU achieves 0.6231 (-7.6%), a further 1.6% improvement over cross-attention with PE.

The progression reveals a clean decomposition: (1) temporal *ordering*—knowing *when* each frame occurred—is the dominant requirement, contributing 79% of the total gain. Sinusoidal PE, which injects position indices without learned weights, is sufficient to capture this. (2) The GRU's causal *sequential integration*—where each hidden state is a running summary of all preceding frames—provides an incremental further gain. Cross-attention with PE attends to all frames simultaneously; the GRU accumulates a compressed history that may be more compatible with the trajectory decoder's expectation of a single context vector. Geodesic supervision requires temporal context to

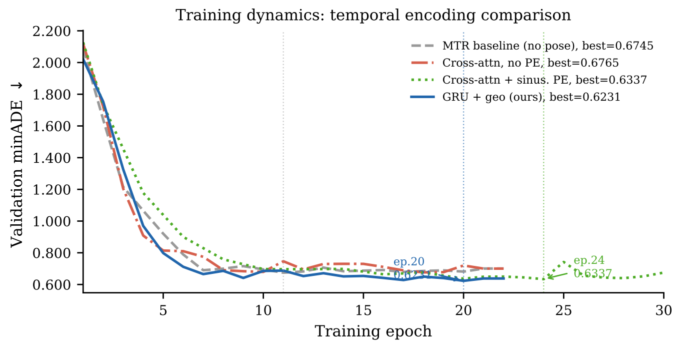

Figure 3: **Training dynamics.** Validation minADE for all four temporal encoding variants. Cross-attention without PE (orange, dash-dot) tracks the baseline throughout, confirming that permutation-invariant aggregation extracts no temporal information. Cross-attention with sinusoidal PE (green, dotted) drops to 0.6337 at epoch 24, recovering 79% of the GRU's advantage. GRU+geo (blue, solid) reaches 0.6231 at epoch 20 with higher epoch-to-epoch variance, consistent with the additional SMPL pose matching signal.

shape encoder features—without ordering (bag-of-items), the geodesic gradient cannot encode gait dynamics.

4.6 Ablation: Future Pose Supervision Quality

To localize the source of improvement, we compare two data conditions: (a) *no future poses*: future pose GT is zero, only past poses are observed (our main experimental condition); and (b) *30fps AMASS future poses*: AMASS motion sequences provide 99% coverage of future timesteps as pseudo-GT.

The result: the 30fps model achieves minADE $= 0.6567$, and the zero-future model achieves 0.6532—a difference of 0.0035 (0.5%), within noise. **Better future pose supervision does not improve trajectory prediction**. This localizes the entire 7.6% gain to the past GRU encoder: encoding how the body has moved up to the present provides useful trajectory context; predicting where the body will go provides no additional benefit.

4.7 Training Dynamics

Figure 3 shows validation minADE over 30 training epochs for all four temporal encoding variants. Cross-attention without PE (orange) closely tracks the trajectory-only baseline, visually confirming the null result. Adding sinusoidal PE (green) produces a qualitatively different trajectory, eventually reaching 0.6337 at epoch 24. MTR+Pose with GRU+geo (blue) reaches its best of 0.6231 at epoch 20 with higher epoch-to-epoch variance, consistent with the additional noise from the SMPL pose supervision signal.

4.8 Ablation: HD Map Context

The experiments above use zero-padded map features (the SMPL-paired pedestrian subset was originally processed without HD map polylines). To quantify how map context interacts with pose conditioning, we enable HD map loading (`WITHOUT_HDMAP: False`) and train the 2×2 factorial design: ±map × ±pose (GRU+geo at $w = 0.1$).

Table 4 reveals two findings. *First*, HD map context is the dominant signal: adding map features alone (no pose) reduces minADE by **27.7%**, far exceeding the 7.6% gain from pose alone. Pedestrian routing—which paths are available, which crosswalks are nearby—is encoded directly in the map, and this spatial prior substantially reduces prediction uncertainty. *Second*, when map context is available, the incremental benefit of pose conditioning shrinks from 7.6% to **1.7%** (0.4880 → 0.4797, still the new overall best). We attribute this to partial information redundancy: body orientation encodes where the pedestrian is heading, which the map's routing constraints already constrain. Without a

Table 4: Map $\times$ pose ablation. All runs: 30 epochs, same training subset. Δ is relative change vs. the no-pose, no-map baseline (0.6745).

Pose Enc.	Map	minADE $\downarrow$	Δ vs. baseline
None	No map	0.6745	—
None	With map	0.4880	-27.7%
GRU+geo	No map	0.6231	-7.6%
GRU+geo	With map	**0.4797**	-28.9%

map, pose fills the gap by providing an orientation proxy; with a map, most of that gap is already closed, leaving only 1.7% residual gain for pose.

5 Analysis and Insights

Why does geodesic loss outperform MPJPE? SMPL joint positions are computed via a nonlinear kinematic chain: $\mathbf{x}_j = \text{FK}(\theta_1, \ldots, \theta_j)$, where $\theta_k \in SO(3)$. Gradient of $\mathcal{L}_{\text{mpjpe}}$ w.r.t. the GRU hidden state traverses this kinematic chain, introducing Jacobians that can be small or misaligned with the gait-relevant rotation axes. Geodesic loss operates directly on $\text{tr}(\mathbf{R}^\top \hat{\mathbf{R}})$, a smooth function of rotation matrices, providing gradients that are proportional to the angular error without kinematic compounding.

What makes the GRU encoder work? The temporal ablation decomposes the GRU's advantage into two separable components. *First*, temporal ordering—knowing *when* each frame occurred—is the dominant requirement. Sinusoidal positional encoding, a fixed injection of position indices requiring no learned parameters, is sufficient to recover 79% of the GRU's gain (-6.0% vs. -7.6%). This is consistent with gait being primarily a phase-ordered phenomenon: knowing that a left-foot strike at $t = 3$ follows right-foot strike at $t = 1$ captures most of the useful temporal structure. *Second*, the GRU's causal *sequential integration* adds an incremental 1.6%. The GRU hidden state $\mathbf{h}_T$ is a compressed running summary of all preceding frames; cross-attention with PE attends to all 11 frames simultaneously and may produce a representation less compatible with the trajectory decoder's expectation of a single compact context vector. With geodesic supervision shaping $\mathbf{h}_T$ toward rotation tracking, the sequential accumulation captures gait phase transitions more efficiently than global self-attention. Importantly, the GRU and cross-attention+PE encoders have nearly identical parameter counts ($\approx$505K vs. $\approx$497K respectively, since sinusoidal PE is parameter-free). The GRU's advantage is therefore purely an inductive bias effect, not a capacity effect.

The mechanism: past encoder, not future prediction. The null result of our future-pose ablation (Section 4.6) is strong evidence that the mechanism is *pose encoding*, not *pose prediction*. The GRU acts as a feature extractor for the MTR decoder, providing an orientation-aware agent representation that improves trajectory intention localization. The pose decoder (future prediction head) is an auxiliary task that supervises the encoder but does not itself contribute to trajectory accuracy.

6 Limitations

Dataset scale. Our experiments use 579 training scenes from Waymo, a small fraction of the full dataset. While relative comparisons are internally valid, we cannot claim that the 7.6% improvement transfers directly to a full-scale training regime.

SMPL pose availability. We use AMASS-derived pose sequences matched to Waymo pedestrian tracks, which requires a separate pose estimation step at test time. In deployment, online body pose estimation (e.g., from video) introduces estimation noise not present in our evaluation. The sensitivity of our method to noisy pose inputs is not evaluated here.

Map context reduces pose benefit. Our main ablations (Sections 4.2–4.5) use zero-padded map features. A follow-up 2$\times$2 ablation (Section 4.8) shows that HD map context alone yields -27.7%

minADE, dwarfing the -7.6% pose gain. With map, pose adds only 1.7% on top. While pose conditioning remains beneficial in map-equipped settings, the relative contribution is substantially smaller—a practitioner deploying MTR+POSE in a full map-enabled system should expect modest, not large, gains from pose conditioning.

Pose decoder quality. We do not evaluate MPJPE/geodesic error of the predicted future poses independently; our metric is trajectory prediction only. A joint evaluation of pose and trajectory quality would complete the picture.

Single-seed results. All 14 runs use a single random seed. Epoch-to-epoch variance in our training curves (particularly the LR-decay spike at epoch 25 in the H5b run, where minADE jumps to 0.7420 before recovering) suggests non-trivial training noise. We report best-checkpoint minADE, which may slightly overfit to the noise of a single run. Repeating key runs with multiple seeds would tighten the confidence interval on the reported improvements.

7 CONCLUSION

We have presented MTR+POSE, a pose-conditioned extension of the Motion Transformer that incorporates past SMPL body poses via a GRU encoder with 6D rotation representations. Through a systematic ablation of 16 runs on the Waymo Open Motion Dataset, we establish that: (1) geodesic supervision in $SO(3)$ is the critical ingredient, reducing minADE by 7.6% over the trajectory-only baseline, while joint-space MPJPE supervision yields only 2.5% improvement and degrades performance when combined with geodesic loss; (2) temporal ordering of pose frames is the dominant requirement—cross-attention *without* positional encoding provides no improvement (0.6765), while adding sinusoidal PE recovers 79% of the GRU's advantage (0.6337, -6.0%); the GRU's causal sequential integration provides a further incremental gain to -7.6%; (3) the source of improvement is the past pose encoder, not future pose prediction quality; (4) HD map context is the dominant signal (-27.7% minADE without any pose), and when map features are available the incremental pose benefit shrinks from 7.6% to 1.7%, revealing partial redundancy between body orientation and map-encoded routing constraints.

These findings suggest that rotation-space supervision and temporal ordering are underappreciated design choices in pose-conditioned forecasting, and that the value of pose conditioning is modulated by available map context. We hope the systematic ablation methodology contributes to more principled design of multi-modal pedestrian prediction systems.

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LLM DISCLOSURE

The authors used large language model (LLM) assistance (Claude, Anthropic) during the preparation of this manuscript for writing drafts, code assistance, and literature search support. All experimental results, scientific analysis, and conclusions are the authors' own.

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A Additional Experimental Details

A.1 Architecture Hyperparameters

- GRU hidden size: 256
- Pose input dimension: 144 (24 joints $\times$ 6D rotation)
- Past pose window: 11 timesteps (1.0 second at 10 Hz)
- Number of predicted trajectory modes: $K = 6$
- Number of future steps: 80 (8.0 seconds)
- MTR encoder: 4-layer transformer, 256 hidden dim, 8 heads
- Cross-attention pose encoder (H5/H5b): pose projection 144→256, MultiheadAttention(embed_dim=256, num_heads=8, dropout=0.1, batch_first=True); $\approx$497K parameters total including fusion MLP
- GRU pose encoder: GRU(input=144, hidden=256, layers=1); $\approx$505K parameters total including fusion MLP
- Batch size: 10
- Optimizer: AdamW, $\beta_1 = 0.9$, $\beta_2 = 0.999$
- LR schedule: Cosine decay, initial LR $1{\times}10^{-3}$
- Epochs: 30
- Gradient clipping: norm ≤ 1.0
- Dropout: none (small dataset)

A.2 Dataset Construction

Waymo agent future trajectories are stored in `processed_scenarios/` PKL files with shape `(num_agents, 91, 10)`. SMPL pedestrian files are matched to Waymo object IDs via integer stem matching. Approximately 80% of validation pedestrians have a valid trajectory match; unmatched pedestrians are excluded from evaluation. The 91-step grid spans $[0, 9.0]$ seconds; SMPL timestamps (originally in microseconds) are divided by 10^6 before alignment.

A.3 Loss Weight Stability

At $w = 1.0$, gradient explosion through the SMPL forward kinematics causes NaN divergence at epoch 3. The geodesic loss gradient norm was also found to diverge at the ± 1 boundary of the $\arccos$ argument; we clamp to $[-1 + \epsilon, 1 - \epsilon]$ with $\epsilon = 10^{-7}$. At $w = 0.1$, training is stable for all 30 epochs across all ablation variants.

A.4 Full Results Table

Table 5 reports all 16 experimental runs including invalid (zero future GT) baselines for completeness.

Table 5: Complete experimental results. Runs 001–002 are invalid (zero future GT) and excluded from analysis. Baseline for Δ is run 004 (no pose, no map).

Run	Tag	Encoder	Losses / Map (w)	minADE	Δ
001	H1 full pose	GRU	all (1.0)	0.6114	*invalid*
002	H2 no pose	—	traj only	0.7724	*invalid*
003	H1_v2 full	GRU	all (0.1)	0.6532	−3.2%
004	H2_v2 baseline	—	traj only	0.6745	—
005	H1 w=0.05	GRU	all (0.05)	0.6557	−2.8%
006	H1 w=0.2	GRU	all (0.2)	0.6542	−3.0%
007	H1_v3 30fps	GRU	all (0.1), 30fps fut	0.6567	−2.6%
008	H3 mpjpe only	GRU	mpjpe+gmm (0.1)	0.6576	−2.5%
009	H3 gmm only	GRU	gmm (0.1)	0.6467	−4.1%
010	H3 geo only	GRU	geo+gmm (0.1)	0.6231	−7.6%
011	H3 geo w=0.05	GRU	geo+gmm (0.05)	0.6786	−1.5%
012	H3 geo w=0.2	GRU	geo+gmm (0.2)	0.6660	−1.3%
013	H5 cross-attn	XAttn	geo+gmm (0.1)	0.6765	+0.3%
014	H5b xattn+PE	XAttn+PE	geo+gmm (0.1)	0.6337	−6.0%
015	H6 no-pose+map	—	traj only + map	0.4880	−27.7%
016	H6 geo+map	GRU	geo+gmm (0.1) + map	**0.4797**	−28.9%